As well as preventing unauthenticated requests consuming your own resources, you should also make sure that you do not respond to unauthenticated requests with more data than they sent to you. Many DoS attacks rely on some form of amplification so that an unauthenticated request to one API results in a much larger response that can be directed at the real target. A popular example are *DNS amplification attacks*, which take advantage of the unauthenticated Domain Name System (DNS) that maps host and domain names into IP addresses. By spoofing the return address for a DNS query, an attacker can trick the DNS server into flooding the victim with responses to DNS requests that they never sent. If enough DNS servers can be recruited into the attack, then a very large amount of traffic can be generated from a much smaller amount of request traffic, as shown in figure 3.2. By sending requests from a network of compromised machines (known as a *botnet*), the attacker can generate very large amounts of traffic to the victim at very little cost to themselves. Amplification attacks are a particular concern in protocols based on UDP (User Datagram Protocol), which are popular in the Internet of Things (IoT). Securing IoT APIs is covered in chapters 12 and 13.

